FIITJEE

ALL INDIA TEST SERIES

FULL TEST – VII

JEE (Advanced)-2019

PAPER – 2

Time Allotted: 3 Hours **Maximum Marks: 180**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-D.**

1. ***Section-A (01 – 06, 19 – 24, 37 – 42)*** *contains 18 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for all correct answer.*
   *Partial Marks : **+3** If all the four options are correct but ONLY three options are chosen.*
   *Partial Marks : **+2** If three or more options are correct but ONLY two options are chosen, both of which are correct options.*
   *Partial Marks : **+1** If two or more options are correct but ONLY one option is chosen and it is a correct option.*
   *Zero Marks : **0** If none of the options is chosen (i.e. the question is unanswered).*
   *Negative Marks : **–2** In all other cases.*

   Section-A (07 – 10, 25 – 28, 43 – 46) *contains 12 questions. Each question has* ***TWO (02)*** *matching lists:* ***LIST-I*** *and* ***LIST-II****.* ***FOUR*** *options are given representing matching of elements from* ***LIST-I*** *and* ***LIST-II****.* ***ONLY ONE*** *of these four options corresponds to a correct matching.*
   For each question, choose the option corresponding to the correct matching.
   Each question has ***only one correct*** *answer and carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

2. ***Section-D (11 – 18, 29 – 36, 47 – 54)*** *contains 24 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

Physics

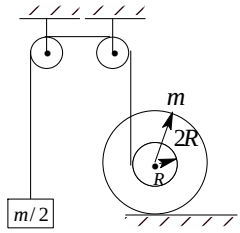
PART – I

SECTION – A

(One or More than one correct type)

This section contains **6** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

1. System shown in the figure is released from rest and spool does not slip at any point of its motion. Thread and pulley are ideal, thread is of sufficient length. Which of following are correct statements after the spool and block are released:



(A) Spool will keep moving toward right
(B) Spool will leave contact with table after covering certain distance
(C) Block will be at rest when spool will have maximum velocity.
(D) Block and spool will both execute oscillatory motion and their ratio of time period will be 2:1

2. A non- conducting disc of radius $1m$ has charge distributed over its surface. Surface charge density at a point is given by $\sigma = A - Br, 0 \le r \le 1$ r is distance from centre. Disc is rotated with angular velocity ω about an axis passing through centre and perpendicular to disc then:

(A) Magnetic field at centre is zero if $\frac{A}{B} = \frac{4}{5}$

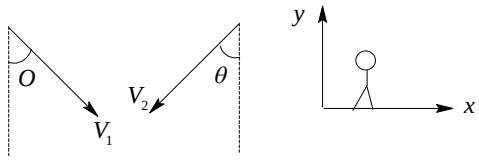
(B) Magnetic field at centre is zero $\frac{A}{B} = \frac{1}{2}$

(C) Magnetic moment of disc is zero if $\frac{A}{B} = \frac{1}{2}$

(D) Magnetic moment of disc is zero if $\frac{A}{B} = \frac{4}{5}$

Space for Rough work

3. V_1 and V_2 are velocity vectors and θ is angle with vertical



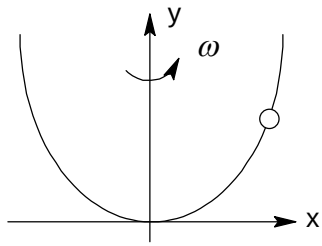
(A) A man moving in positive x direction observes rain in direction of V_1 vector, if it moves with increasing speed it will observe that rain velocity vector is rotating clockwise

(B) A man moving in positive x direction observes rain in direction of V_2 vector, if it moves with decreasing speed it will observe that rain velocity vector is rotating anti-clockwise

(C) A man moving in negative x direction observes rain in direction of V_1 vector, if it moves with decreasing speed it will observe that rain velocity vector is rotating anti-clockwise

(D) A man moving in negative x direction observes rain in direction of V_2 vector, if it moves with increasing speed it will observe that rain velocity vector is rotating anti anti-clockwise

4. A curved smooth wire as shown in diagram is revolving with constant angular velocity about y-axis, a bead in this wire remain in equilibrium at every position then shape of wire:



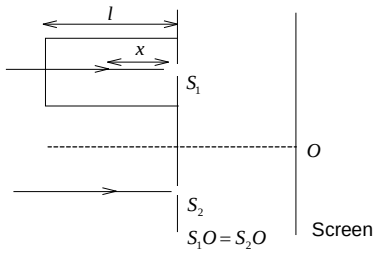
(A) May be straight line

(B) Parabolic

(C) $y \propto x^4$

(D) Not possible for any shape

5. In the figure shown, a parallel beam of light is incident on the plane of the slits of a Young's double slit experiment. Light incident on the slit, S, passes through a medium of variable refractive index $\mu = 1 + ax$ (where x is distance from the plane of slits as shown) upto a distance 'l' before falling on S_1. Rest of the space is filled with air. If at 'O' a minima is formed, then the minimum value of the positive constant a is:



(A) $\frac{\lambda}{l}$ (B) $\frac{\lambda}{l^2}$

(C) $\frac{l^2}{\lambda}$ (D) $\frac{l^2}{2\lambda}$

Space for Rough work

6. The photon radiated from hydrogen corresponding to 2nd line of Lyman series is absorbed by a hydrogen like atom x in 2nd excited state. As a result the hydrogen like atom 'x' makes a transition to nth orbit. Then

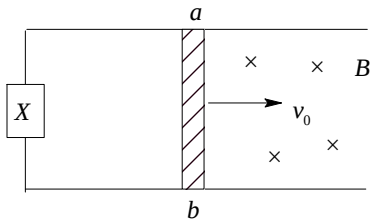
(A) $x = He^{+}, n = 4$ (B) $x = Li^{++}, n = 6$

(C) $x = He^{+}, n = 6$ (D) $x = Li^{++}, n = 9$

(Matching List Type)

This section contains **FOUR** questions. Each question has **TWO** matching lists: **LIST–I** and **LIST–II**. **FOUR** options are given representing matching of elements from **LIST–I** and **LIST–II**. **ONLY ONE** of these four options corresponds to a correct matching.

7. A conducting rod shown in figure of mass m and length ℓ moves on two frictionless parallel rails in the presence of uniform magnetic field directed into the page. X is the load connected between parallel rails. The rod is given a velocity v_0 to the right and is released at $t = 0$ match List– I and II:

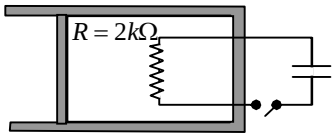


List – I		List – II	
(P)	If X is a resistor	(1)	The rod moves under action of magnetic force
(Q)	X is ideal inductor	(2)	Energy is consumed
(R)	If X is a capacitor and rod is dragged by a constant force	(3)	velocity of rod increases linearly with time
(S)	X is a capacitor and arrangement are fixed vertically and conductor is released from rest	(4)	rod oscillate simple harmonically

(A) (P) $\rightarrow$(1, 2), (Q) $\rightarrow$(1, 2, 4), (R) $\rightarrow$(1, 2, 3), (S) $\rightarrow$ (1, 2, 3)
(B) (P) $\rightarrow$(2, 1), (Q) $\rightarrow$(4, 3, 1), (R) $\rightarrow$(3, 4, 1), (S) $\rightarrow$(2, 3, 4)
(C) (P) $\rightarrow$(3, 4), (Q) $\rightarrow$ (1, 3, 4), (R) $\rightarrow$(3, 2, 4), (S) $\rightarrow$ (1, 3, 4)
(D) (P) $\rightarrow$(3, 1), (Q) $\rightarrow$ (2, 4, 3), (R) $\rightarrow$(4, 1, 2), (S) $\rightarrow$(1, 2, 4)

Space for Rough work

8. A cylinder fitted with a piston, which can slide without friction contains one mole of an ideal gas. The walls of the cylinder and the piston are adiabatic. The cylinder contains a resistor of resistance R = 2.0 kΩ which is connected to a capacitor of capacity C = 75mF. Initially, potential difference across capacitor is 640/3 volt and the switch is open. When the switch is closed for time =2.5 ln 4 minutes, the gas expands isobarically and its temperature changes by 72k. Heat loss through the wires is negligible.



List – I		List – II	
(P)	The work done by the gas (in K J)	(1)	1.6
(Q)	Increment in internal energy of the gas (in K J)	(2)	1
(R)	The value of γ	(3)	0.6
(S)	Energy supplied to the gas (in K J)	(4)	1.3

(A) (P) $\rightarrow$(3), (Q) $\rightarrow$(2), (R) $\rightarrow$(1), (S) $\rightarrow$(1)
(B) (P) $\rightarrow$(2), (Q) $\rightarrow$(1), (R) $\rightarrow$(3), (S) $\rightarrow$(1)
(C) (P) $\rightarrow$(1), (Q)$\rightarrow$(2), (R) $\rightarrow$(4), (S) $\rightarrow$(1)
(D) (P) $\rightarrow$(2), (Q) $\rightarrow$(2), (R) $\rightarrow$(1), (S) $\rightarrow$(4)

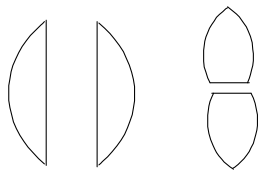
9. A ray of light falls normally on an equilateral prism of refractive index $\mu = \sqrt{3}$. Match the following table:
i_1 = Angle of incidence
i_2 = Angle of emergence from second surface of prism
r_1 = Angle of refraction at the first surface of prism
r_2 = Angle of incidence at second surface of prism

List – I		List – II	
(P)	Angle i_1	(1)	0°
(Q)	Angle r_1	(2)	90°
(R)	Angle i_2	(3)	60°
(S)	Angle r_2	(4)	None

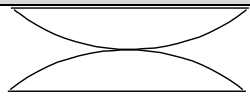
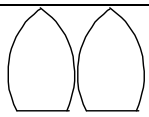
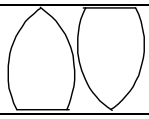
(A) (P) $\rightarrow$(4), (Q) $\rightarrow$(3), (R) $\rightarrow$(2), (S) $\rightarrow$(1)
(B) (P) $\rightarrow$(3), (Q) $\rightarrow$(2), (R) $\rightarrow$(1), (S) $\rightarrow$(2)
(C) (P) $\rightarrow$(2), (Q) $\rightarrow$(1), (R) $\rightarrow$(3), (S) $\rightarrow$(4)
(D) (P)$\rightarrow$(1), (Q) $\rightarrow$(1), (R) $\rightarrow$ (4), (S) $\rightarrow$(3)

Space for Rough work

10. A thin biconvex lens of small aperture and having focal length 30 cm is cut tow ways as shown in the figure.



Match the focal length of different combination given in list–I with their value in list–II

List – I		List – II	
(P)		(1)	Infinite
(Q)		(2)	30 cm
(R)		(3)	60 cm
(S)		(4)	15 cm

(A) (P) $\rightarrow$(3), (Q) $\rightarrow$(2), (R) $\rightarrow$(1), (S) $\rightarrow$(4)
(B) (P) $\rightarrow$(4), (Q) $\rightarrow$(1), (R) $\rightarrow$(2), (S) $\rightarrow$(3)
(C) (P) $\rightarrow$(2), (Q) $\rightarrow$(3), (R) $\rightarrow$(4), (S) $\rightarrow$(1)
(D) (P) $\rightarrow$(1), (Q) $\rightarrow$(4), (R) $\rightarrow$(3), (S) $\rightarrow$(2)

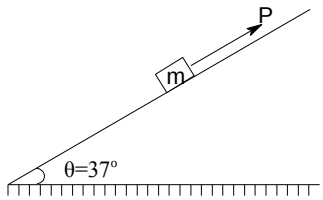
Space for Rough work

SECTION – D
(Numerical Answer Type)

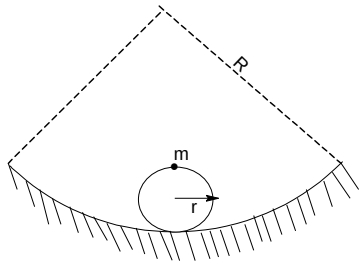
This section contains **EIGHT** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. xxxxx.xx).

11. Steel has density $7800\,\text{kg/m}^3$ and will break if tensile stress exceeds $7.0\times10^8\ \text{N/m}^2$. A person wants to make a guitar string from 4.0 gm of this steel. The guitar string must be able to withstand a tension of 900 N. The maximum speed of transverse wave can be $\text{x.xx}\times100$ m/s

12. A block of mass m is being pulled up the rough incline, inclined at an angle 37° with horizontal by an agent delivering constant power P. The coefficient of friction between the block and the incline is μ. Find the maximum speed (in km/hr) of the block during the course of ascent. [Take : P = 60 W, m = 1 kg, μ = 0.5]



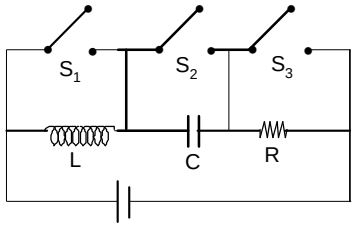
13. A ring of radius r and mass 2m, rests on a rough circular track of radius R.A small point mass of mass m is mounted on its periphery. If friction is sufficient to avoid any slipping then the minimum ratio of $\frac{r}{R}$ for which the system oscillates for small disturbances is



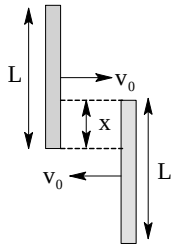
14. The internal energy of a monatomic ideal gas is 1.5 nRT. One mole of helium is kept in a cylinder of cross-section 8.5 cm^2. The cylinder is closed by a light frictionless piston. The gas is heated slowly in a process during which a total of 42 J heat is given to the gas. The temperature rises through 2°C. The distance moved by the piston is given as $a \times 10^b$ m in scientific notation. Find the value of a + b. [Take R=25/3 in SI unit, atmospheric pressure = 100 kPa]

Space for Rough work

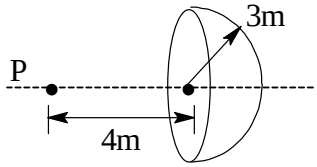
15. Consider the circuit shown in figure. With switch S_1 closed and the other two switches open, the circuit has a time constant 0.05 sec. With switch S_2 closed and the other two switches open, the circuit has a time constant 2 sec. With switch S_3 closed and the other two switches open, the circuit oscillates with a period T. Find T (in sec). (Take π^2 = 10)



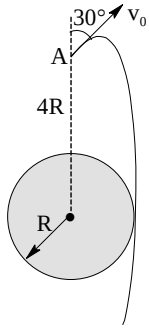
16. Two identical rigid rods, each of mass 1kg & length $\sqrt{3}$m, are moving opposite to each other without rotation on a smooth horizontal table as shown. For what maximum value of 'x' for which the direction of motion of each rod would not change after collision, irrespective of the type of collision. Assume no sticking.



17. A point source of power 'P' is placed on the axis of a thin hemispherical shell as shown. The shell has a radius of 3m and behaves like a perfect black body. If the steady state temperature of shell is T. Find T^4. (Take $P=2400\pi\sigma$ W, σ=stefan's const.).



18. A particle is projected from point 'A', that is at a distance $4R$ from the centre of the Earth, with speed v_0 as shown. If the particle passes grazing the earth's surface, then it is found that $v_0=1000\times N$. Find the value of N [Take $\frac{GM}{R}=6.4\times10^7\ m^2/s^2$]



Space for Rough work

Chemistry

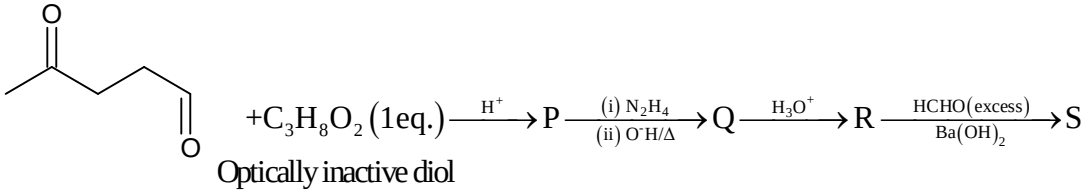
PART – II

SECTION – A

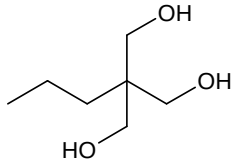
(One or More than one correct type)

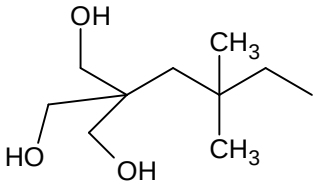
This section contains **6** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

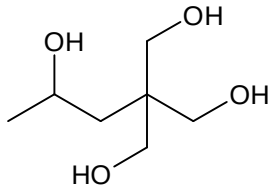
19. Final product 'S' is:



(A) Compound 'S' is triol

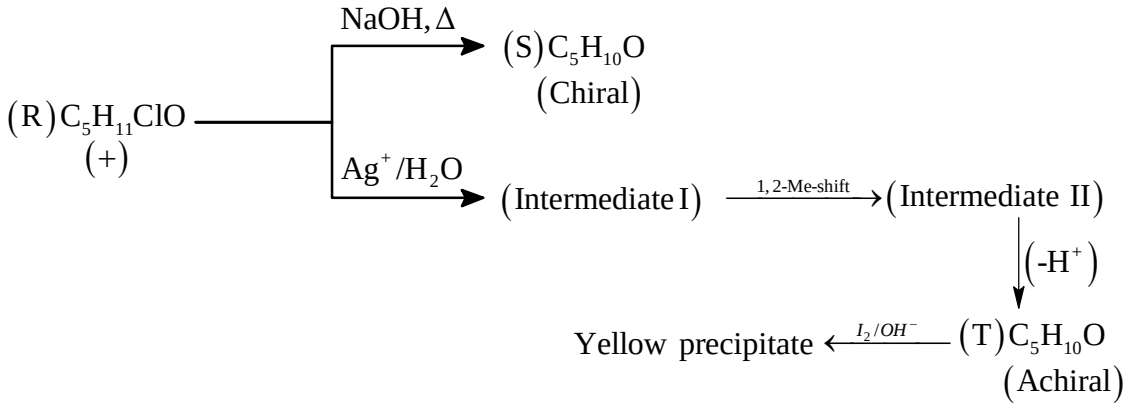
(B) 

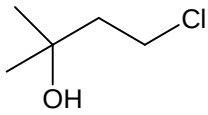
(C) 

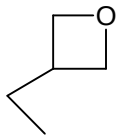
(D) 

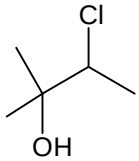
Space for Rough work

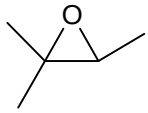
20. The structures of the compounds R and S are:



(A) 

(B) 

(C) 

(D) 

Space for Rough work

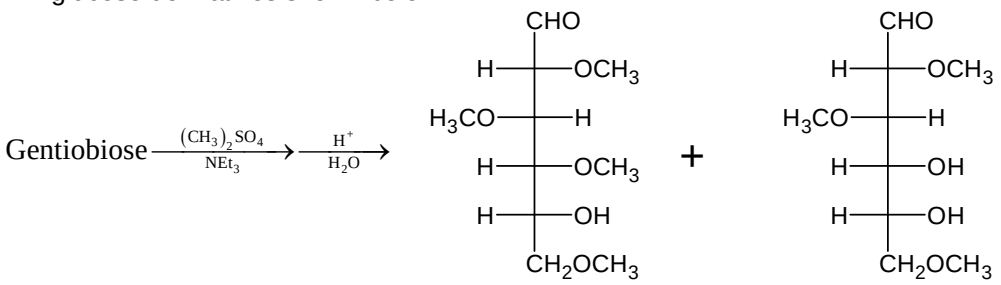
21. Gentiobiose is a disaccharide incorporated into the chemical structure of crocin, the chemical compound that gives saffron its colour. Deduce the structure of gentiobiose from the following information.

(i) Acid hydrolysis of gentiobiose yields only D-glucose

(ii) Gentiobiose is hydrolysed by β-glucosidase enzymes (breaks β-glucosidic linkage) , but not by $\alpha-$glucosidase enzymes

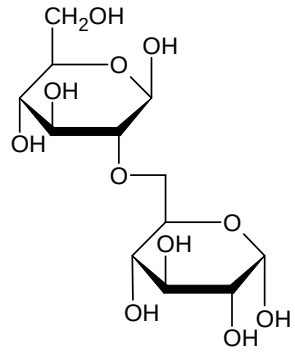
(iii) Gentiobiose is a reducing sugar and reacts with sodium borohydride

(iv) Methylation of gentiobiose followed by hydrolysis of the glycosidic bonds yields the two D-glucose derivatives shown below.

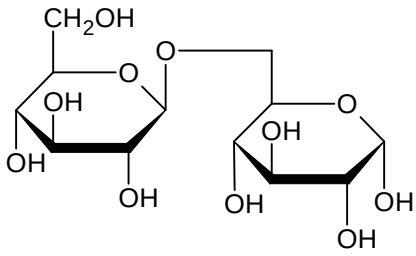


Which can't be the structure of gentiobiose?

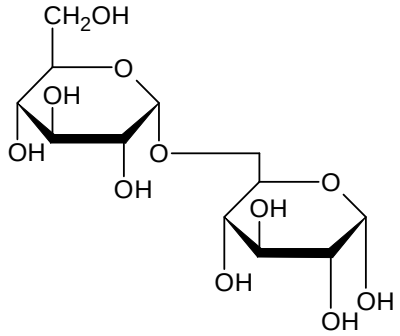
(A)



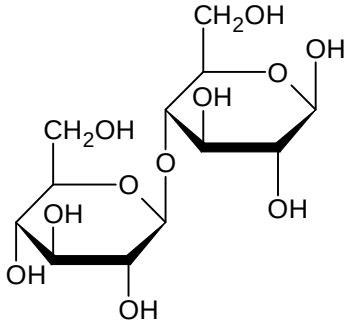
(B)



(C)



(D)



Space for Rough work

22. Which of the following is/are not lyophilic sol?
(A) Gelatin (B) Silver sol
(C) Sulphur sol (D) A_2S_3 sol

23. A 10 mL mixture containing 0.2 g NaOH and $0.25\,g\ Na_2CO_3$ was treated with $x N\ HCl$ required 15 mL of it with phenolphthalein indicator. After end point a $0.5\ g\ NaHCO_3$ was added and it was again titrated with $y\ N\ HCl$ required 25.5 mL of HCl with methyl orange. Which of the following is/are correct?
(A) x is almost 1.5 times of y
(B) If in above titration x N HCl is replaced with y N HCl then for first end point it will require roughly 23mL of HCl.
(C) After first end point instead of 0.5 g if 2.5 g $NaHCO_3$ is added and fitrated with $x\ N\ HCl$ it will require 65.5 mL.
(D) Only (A) and (B) are correct.

24. In which of the following pair the later species is not having higher vapour pressure at a given temperature than former one?
(A) C_2H_6 and C_4H_{10} (B) CH_3CH_2OH and CH_3CH_2F
(C) Xe and I_2 (D) NH_3 and PH_3

(Matching List Type)

This section contains **FOUR** questions. Each question has **TWO** matching lists: **LIST–I** and **LIST–II**. **FOUR** options are given representing matching of elements from **LIST–I** and **LIST–II**. **ONLY ONE** of these four options corresponds to a correct matching.

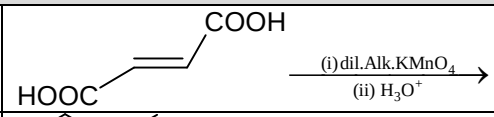
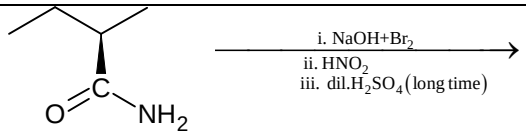
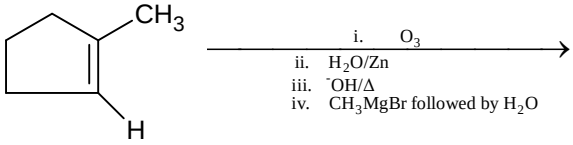
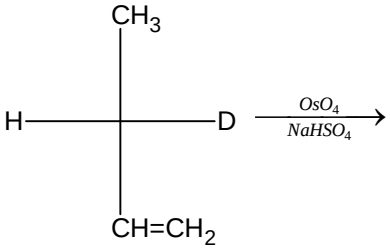
25. Match the list:

List-I		List-II	
(P)	$ZnCO_3 \rightarrow Zn$	(1)	Calcination
(Q)	$ZnS \rightarrow ZnO$	(2)	Roasting
(R)	$PbS\left(high\ grade\right) \rightarrow Pb$	(3)	Self reduction
(S)	$Cu_2S \rightarrow Cu_2O$	(4)	Carbon reduction
		(5)	No change in oxidation state of metal

(A) (P) $\rightarrow$ (1), (Q) $\rightarrow$ (2), (R) $\rightarrow$ (3), (S) $\rightarrow$ (5)
(B) (P) $\rightarrow$ (1), (Q) $\rightarrow$ (4), (R) $\rightarrow$ (3), (S) $\rightarrow$ (5)
(C) (P) $\rightarrow$ (3), (Q) $\rightarrow$ (2), (R) $\rightarrow$ (2), (S) $\rightarrow$ (2)
(D) (P) $\rightarrow$ (4), (Q) $\rightarrow$ (2), (R) $\rightarrow$ (3), (S) $\rightarrow$ (1)

Space for Rough work

26. Match the list-

List-I		List-II	
(P)		(1)	The product (s) consist two chiral centre
(Q)		(2)	The products are diastereomers
(R)		(3)	The product is racemic mixture
(S)		(4)	The product consist single chiral centre
		(5)	The Product does not consist chiral centre.

(A) (P) $\rightarrow$ (1), (Q) $\rightarrow$ (3), (R) $\rightarrow$ (4), (S) $\rightarrow$ (2)
(B) (P) $\rightarrow$ (3), (Q) $\rightarrow$ (4), (R) $\rightarrow$ (5), (S) $\rightarrow$ (2)
(C) (P) $\rightarrow$ (1), (Q) $\rightarrow$ (2), (R) $\rightarrow$ (4), (S) $\rightarrow$ (1)
(D) (P) $\rightarrow$ (5), (Q) $\rightarrow$ (3), (R) $\rightarrow$ (4), (S) $\rightarrow$ (1)

Space for Rough work

27. Match the conditions on left list with solution on right list:

List-I		List-II	
(P)	Has Maximum pH at the end point when titrated against KOH	(1)	$CH_3COOH\left(K_4 = 2\times10^{-5}\right)$
(Q)	Has minimum pH at the end point when titrated against KOH	(2)	$HCN\left(K_2 = 1\times10^{-10}\right)$
(R)	Evolve maximum heat when treated with strong acid	(3)	$CH_3COO^-Na^+$
(S)	Evolve minimum when treated with strong acid	(4)	$NH_3\left(K_b = 2\times10^{-5}\right)$
		(5)	$NaCN$

(A) (P) $\rightarrow$ (2), (Q) $\rightarrow$ (1), (R) $\rightarrow$ (5), (S) $\rightarrow$ (3)
(B) (P) $\rightarrow$ (2), (Q) $\rightarrow$ (1), (R) $\rightarrow$ (3), (S) $\rightarrow$ (5)
(C) (P) $\rightarrow$ (1), (Q) $\rightarrow$ (2), (R) $\rightarrow$ (3), (S) $\rightarrow$ (5)
(D) (P) $\rightarrow$ (1), (Q) $\rightarrow$ (2), (R) $\rightarrow$ (5), (S) $\rightarrow$ (3)

28. Match the list-

List-I (Radical which can decolorize acidic solution of MnO_4^-)		List-II (Given salt)	
(P)	Only acidic radical	(1)	$Sn\left(NO_3\right)_2$
(Q)	Only basic radical	(2)	KNO_2
(R)	Both radical	(3)	FeC_2O_4
(S)	Neither acidic nor basic radical	(4)	Na_2CO_3
		(5)	$BaSO_3$

(A) (P) $\rightarrow$ (2), (Q) $\rightarrow$ (1), (R) $\rightarrow$ (3), (S) $\rightarrow$ (4)
(B) (P) $\rightarrow$ (1), (Q) $\rightarrow$ (2), (R) $\rightarrow$ (3), (S) $\rightarrow$ (4)
(C) (P) $\rightarrow$ (5), (Q) $\rightarrow$ (2), (R) $\rightarrow$ (3), (S) $\rightarrow$ (4)
(D) (P) $\rightarrow$ (2), (Q) $\rightarrow$ (1), (R) $\rightarrow$ (4), (S) $\rightarrow$ (3)

Space for Rough work

SECTION – D
(Numerical Answer Type)

This section contains **EIGHT** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. xxxxx.xx).

29. A reaction $2A+B\longrightarrow C+2D$ is first order with respect to A and second order with respect to B. Initial concentration of A is 'x' while that of B is '2x'. Initial concentration of C is zero. If at t = 30 minutes, the concentration of C is x/4, the rate expression for decomposition of A at t = 30 minutes is 'rate = $\frac{49x^3kZ}{16}$. (Where 'k' is the overall rate constant) The value of Z is:

30. If 10 g V_2O_5 is dissolved in acid and reduced to V^{2+} by treatment with zinc metal, the mole of I_2 could be reduced by the resulting V^{2+} solution, as it is oxidised to V^{4+}*is* x, then the value of 'x' *will be*? $\left(\text{At. wt. of V}=50\right)$

31. A solution contains a mixture of $\text{Ag}^+\left(0.10\text{M}\right)$ and Hg_2^{2+} (0.10 M) which is to be separated by selective precipitation by adding iodide ion at which one of them gets precipitated $\text{X}\%$ of one metal ion is precipitated till second one starts to precipitate. The value of X is?
[K_{sp}:AgI = 8.5×10^{-17}, Hg_2I_2 = 2.5×10^{-26}]

32. The Henry law constant for dissolution of a gas in aqueous medium is 3×10^2 atm. At what partial pressure of the gas (in atm), the molality of gas in aqueous solution will be $\frac{5}{9}\text{m}$.

Space for Rough work

33 An alloy of Pb-Ag weighing 54 mg was dissolved in sufficient amount of HNO_3 & volume was made upto 500ml. An Ag electrode was dipped in solution and then connected to standard hydrogen electrode anode. Then % of Ag in alloy will be

[Given: $E_{cell} = 0.5V$; $E^0_{Ag^+/Ag} = 0.8V$, $\frac{2.303RT}{F} = 0.06$,Atomic weight of silver=108]

34. Given that crystal field stabilization energy for $\left[Co(H_2O)_6\right]^{3+}$ is -8800 cm^{-1}, pairing energy for Co^{3+} is $20,800\,cm^{-1}$, then the value of Δ_0 is $'x' \times 10^3\left(\text{in cm}^{-1}\right)$, *then* value of 'x' is :

35. The vapour pressure lowering of a 0.1m aqueous solution of non – electrolyte at $75°C$ is $\left(\Delta H^0_v \text{ of water}=9720\,\text{cal/mol and } e^{-0.936} = 0.3921,\ R=2\text{cal/mol/K}\right)$

36. The C – C single bond distance is $1.5Å$. The minimum distance between the terminal cartons in propane is $\left(\sin 54°45' = 0.8\right)$

Space for Rough work

Mathematics

PART – III

SECTION – A

(One or More than one correct type)

This section contains **6** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

37. If $f(x) = \int_1^x \frac{\ln t}{1+t}dt$, then

(A) $f\left(\frac{1}{x}\right) = -\int_1^x \frac{\ln t}{t(1+t)}dt$

(B) $f\left(\frac{1}{x}\right) = \int_1^x \frac{\ln t}{t(1+t)}dt$

(C) $f(x) + f\left(\frac{1}{x}\right) = 0$

(D) $f(x) + f\left(\frac{1}{x}\right) = \frac{1}{2}(\ln x)^2$

38. The equation $2\sin\frac{x}{2}\cos^2\frac{x}{2} - 2\sin\frac{x}{2}\sin^2 x = \cos^2 x - \sin^2 x$ has a root for which

(A) sin2x = 1

(B) sin2x = – 1

(C) cosx = $\frac{1}{2}$

(D) cos2x = – $\frac{1}{2}$

39. The equation of the circle which touches the axes of coordinates and the line $\frac{x}{3}+\frac{y}{4}=1$ and whose centre lies in the first quadrant is $x^2 + y^2 – 2cx – 2cy + c^2 = 0$ where c is

(A) 1

(B) 2

(C) 3

(D) 6

40. $\sin^{-1}x > \cos^{-1}x$ holds for

(A) all values of x

(B) $x \in \left(0, \frac{1}{\sqrt{2}}\right)$

(C) $x \in \left(\frac{1}{\sqrt{2}}, 1\right)$

(D) x = 0.75

Space for Rough work

41. $2\cot^{-1}7+\cos^{-1}\frac{3}{5}$ is equal to

(A) $\cot^{-1}\left(\frac{44}{117}\right)$ (B) $\operatorname{cosec}^{-1}\left(\frac{125}{117}\right)$

(C) $\tan^{-1}\left(\frac{4}{117}\right)$ (D) $\cos^{-1}\left(\frac{44}{125}\right)$

42. The interval in which this is valid $\cot\frac{\theta}{2}\geq 1+\cot\theta$

(A) $0<\theta<\frac{\pi}{4}$ (B) $\frac{\pi}{4}<\theta<\frac{\pi}{2}$

(C) $0<\theta<\pi$ (D) none of these

(Matching List Type)

This section contains **FOUR** questions. Each question has **TWO** matching lists: **LIST–I** and **LIST–II**. **FOUR** options are given representing matching of elements from **LIST–I** and **LIST–II**. **ONLY ONE** of these four options corresponds to a correct matching.

43. Match the following:

List – I		List – II	
(P)	Domain of $\frac{\log_2(x+3)}{x^2+8x+15}$	(1)	$(-3, \infty)$
(Q)	Domain of $\sqrt{1-\log_3 x}$	(2)	(0, 3]
(R)	Domain of $\frac{\sqrt{4-x^2}}{\sin(2-x)}$	(3)	[1, 2)
(S)	Domain of $\log[1-\log_9(x^2-3x+9)]$	(4)	(0, 3)
		(5)	[–4, 1)

(A) (P) $\rightarrow$ (4), (Q) $\rightarrow$ (3), (R) $\rightarrow$ (2), (S) $\rightarrow$ (1)
(B) (P) $\rightarrow$ (1), (Q) $\rightarrow$ (2), (R) $\rightarrow$ (3), (S) $\rightarrow$ (4)
(C) (P) $\rightarrow$ (4), (Q) $\rightarrow$ (2), (R) $\rightarrow$ (1), (S) $\rightarrow$ (3)
(D) (P) $\rightarrow$ (5), (Q) $\rightarrow$ (2), (R) $\rightarrow$ (4), (S) $\rightarrow$ (1)

Space for Rough work

44. Match the list:

List – I		List – II	
(P)	In a ΔABC if area $\Delta = a^2 - (b-c)^2$, then 15tanA is	(1)	$\frac{1}{3}$
(Q)	In a ΔABC a = 6, b = 3 and $\cos(A-B) = \frac{4}{5}$, then area of ΔABC is	(2)	2
(R)	If a, b, c, d are the sides of quadrilateral, then the minimum value of $\frac{a^2+b^2+c^2}{d^2}$ is	(3)	3
(S)	ΔPRQ is right angled triangle where P(3, 1), Q(6, 5) and R(x, y) and area of ΔPRQ = 7, then number of such point R is	(4)	8
		(5)	9

(A) (P) $\rightarrow$ (4), (Q) $\rightarrow$ (5), (R) $\rightarrow$ (1), (S) $\rightarrow$ (2)
(B) (P) $\rightarrow$ (1), (Q) $\rightarrow$ (5), (R) $\rightarrow$ (1), (S) $\rightarrow$ (1)
(C) (P) $\rightarrow$ (2), (Q) $\rightarrow$ (5), (R) $\rightarrow$ (1), (S) $\rightarrow$ (4)
(D) (P) $\rightarrow$ (1), (Q) $\rightarrow$ (2), (R) $\rightarrow$ (3), (S) $\rightarrow$ (4)

45. If $\int_0^{\infty} \frac{\sin x}{x} dx = \frac{\pi}{2}$, then match the following

List – I		List – II	
(P)	$\int_0^{\infty} \frac{\sin 5x}{x} dx$	(1)	0
(Q)	$\int_0^{\infty} \frac{\sin ax \cos bx}{x} dx$ (a> b > 0)	(2)	$\frac{\pi}{2}$
(R)	$\int_0^{\infty} \frac{\sin^2 x}{x^2} dx$	(3)	$\frac{\pi}{4}$
(S)	$\int_0^{\infty} \frac{\sin^3 x}{x} dx$	(4)	π
		(5)	$\frac{\pi}{6}$

(A) (P) $\rightarrow$ (1), (Q) $\rightarrow$ (2), (R) $\rightarrow$ (3), (S) $\rightarrow$ (4, 5)
(B) (P) $\rightarrow$ (1), (Q) $\rightarrow$ (5), (R) $\rightarrow$ (1), (S) $\rightarrow$ (1, 4)
(C) (P) $\rightarrow$ (2), (Q) $\rightarrow$ (2), (R) $\rightarrow$ (2), (S) $\rightarrow$ (1, 3)
(D) (P) $\rightarrow$ (1), (Q) $\rightarrow$ (2), (R) $\rightarrow$ (3), (S) $\rightarrow$ (2, 4)

Space for Rough work

46. Match the following:

List – I		List – II	
(P)	Lines with direction ratios (1, – c, – b), (– c, – 1, – a) and (–b, – a, 1) are coplanar then $a^2 + b^2 + c^2 + 2abc$ is	(1)	–1
(Q)	If the lines x = ay + 1, x = by + 2 and x = cy + 3, z = dy + 4 are perpendicular then ac + bd is equal to	(2)	1
(R)	If (a, b, c) lies on a plane which form ΔABC with axes whose centroid lies on (α, β, γ) then $\frac{a}{\alpha}+\frac{b}{\beta}+\frac{c}{\gamma}$ is equal to	(3)	3
(S)	Let [.] denotes the greatest integer less than or equal to x, then f(x) = [x sinπx] is not differentiable if x =	(4)	0
		(5)	2

(A) (P) $\rightarrow$ (1), (Q) $\rightarrow$ (2), (R) $\rightarrow$ (3), (S) $\rightarrow$ (4, 5)
(B) (P) $\rightarrow$ (1), (Q) $\rightarrow$ (5), (R) $\rightarrow$ (1), (S) $\rightarrow$ (1, 4)
(C) (P) $\rightarrow$ (2), (Q) $\rightarrow$ (2), (R) $\rightarrow$ (2), (S) $\rightarrow$ (1, 3)
(D) (P) $\rightarrow$ (2), (Q) $\rightarrow$ (1), (R) $\rightarrow$ (3), (S) $\rightarrow$ (1, 2)

SECTION – D
(Numerical Answer Type)

This section contains **EIGHT** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. xxxxx.xx).

47. Square of diameter of the circle having tangent at (1, 1) as x + y – 2 = 0 and passing through (2, 2) is _____

48. In an acute angled triangle the minimum value of secA secB secC(1 + secA)(1 + secB)(1 + secC) is k, then k is __

49. P is any point and O being the origin. On the circle with OP as diameter two point Q and R are on same side of OP such that $\angle$POQ = $\angle$QOR = θ. Let P, Q, R be z_1, z_2, z_3 such that $2\sqrt{3}z_2^2 = (2+\sqrt{3})z_1z_3$. The degree measure of θ is α°, then α is _____

Space for Rough work

50. The number of solutions of $\theta \in [0, 2\pi]$ satisfying the equation $\log_{\sqrt{5}} \tan\theta \sqrt{\log_{\tan\theta} 5\sqrt{5} + \log_{\sqrt{5}} 5\sqrt{5}} = -\sqrt{6}$ is __

51. $I = \int_0^{1.5} \left[x^2\right] dx$ where [.] is greatest integer function then the value of I is _____

52. If $|z|^2 + (3 - 4i)z + (3 + 4i)\bar{z} + 75 = 0$ and $(1 - i)z + (1 + i)\bar{z} - 16 = 0$ intersect at z_1 and z_2, then the sum of the areas of the quadrilaterals having $(z_1 + z_2)$ and $(z_1 - z_2)$ as diagonals passing through origin is k, then k is _____ (Two vertices of 1st quadrilateral are z_1 and z_2 and of 2nd quadrilateral are z_1 and $-z_2$).

53. If α is the absolute maximum value of the expression $\frac{3x^2 + 2x - 1}{x^2 + x + 1}$ $\forall$ x $\in$ R, then $[\alpha]$is ___, (where [.] denotes the greatest integer function)

54. In a triangle ABC, side AB = 20, AC = 11, BC = 13, then the diameter of the semicircle inscribed in triangle ABC, whose diameter lies on AB and is touching AC and BC is d, then d is _____

Space for Rough work